

Design, modeling and kinematic analysis of a multi-configuration dexterous hand with integrated high-dimensional sensors

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Abstract

Purpose – This study aims to introduce a multi-configuration, three-finger dexterous hand with integrated high-dimensional sensors and provides an analysis of its design, modeling and kinematics.

Design/methodology/approach – A mechanical design scheme of the three-finger dexterous hand with a reconfigurable palm is proposed based on the existing research on dexterous hands. The reconfigurable palm design enables the dexterous hand to achieve four grasping modes to adapt to multiple grasping tasks. To further enhance perception, two six-axis force and torque sensors are integrated into each finger. The forward and inverse kinematics equations of the dexterous hand are derived using the D-H method for kinematics modeling, thus providing a theoretical model for index analysis. The performance is evaluated using three widely applied indicators: workspace, interactivity of fingers and manipulability.

Findings – The results of kinematics analysis show that the proposed hand has excellent dexterity. Additionally, three different experiments are conducted based on the proposed hand. The performance of the dexterous hand is also verified by fingertip force, motion accuracy test, grasping and in-hand manipulation experiments based on Feix taxonomy. The results show that the dexterous hand has good grasping ability, reproducing 82% of the natural movement of the human hand in daily grasping activities and achieving in-hand manipulations such as translation and rotation.

Originality/value – A novel three-finger dexterous hand with multi-configuration and integrated high-dimensional sensors is proposed. It performs better than the previously designed dexterous hand in actual experiments and kinematic performance analysis.

Keywords Dexterous hand, Kinematics analysis, Robot grasp, In-hand manipulation

Paper type Research paper

1. Introduction

The robot generally consists of two parts: the manipulator and the end-effector. The end-effector usually includes special tools and general grippers. As a universal end-effector, the dexterous hand can emulate the structure and function of the human hand. The dexterous hand could grasp and manipulate irregular objects in unstructured environments. It has the characteristics of task versatility and wide application range. Currently, application of dexterous hands has been extended to various domains, including aerospace (Lovchik and Diftler, 1999), underwater applications (Bemfica *et al.*, 2014; Wang *et al.*, 2022) and the service industry (Wang *et al.*, 2019), among others. Pizza conducted a comprehensive review of the development of the dexterous hand over the past century (Piazza *et al.*, 2019). The analysis of the review reveals that dexterous hand research has continuously remained a focus area within the field of robotics.

Many researchers have conducted in-depth research on the mechanism design, control systems and sensors of dexterous hands, yielding fruitful results (Kyberd *et al.*, 2001; Liu *et al.*, 2008; Sun *et al.*, 2022). However, the high degree of freedom (DOF) inevitably makes mechanisms and control systems of the dexterous hand more complex; thus, most dexterous hands are still in the laboratory stage. To solve the problem of system redundancy, a current trend is to reduce the number of fingers. The taxonomy of hand gripping (Feix *et al.*, 2016) shows that the most precise gripping and manipulation is performed with three or fewer fingers, specifically the thumb, index finger and middle finger. Salisbury and Roth (1983) concluded that the dexterous hand needs at least three fingers to achieve stable grasping of a 3D object. Therefore, a

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dexterous hand designed with three fingers is an effective solution for performing diverse grasping tasks.

The dexterous hand can be categorized into two distinct types based on the driving mode: underactuated and fully actuated hand. The underactuated dexterous hand (Catalano *et al.*, 2014; Hong *et al.*, 2019; Li *et al.*, 2022) has many advantages. For example, C-shaped anthropomorphic hand (Yan *et al.*, 2022) increases the reverse stiffness at the joint with the simple structure, but retains high dexterity and adaptability. Underactuated hand has fewer actuators, can reduce the weight and size of the dexterous hand and can provide better grasping adaptability. However, underactuated hands are characterized by limited motion accuracy, which poses challenges in achieving stable grasping and manipulation of objects with varying sizes. The fully actuated dexterous hand allows for precise control of its fingers, and each joint can move independently of each other. The use of fully actuated fingers is beneficial to improve the DOF and dexterity of the dexterous hand. Furthermore, while researchers have developed numerous soft dexterous hands (Hughes *et al.*, 2016; Watanabe *et al.*, 2017) using deformable materials to improve dexterity, real-world applications of these systems remain restricted because of current limitations in materials and control strategies. Based on previous studies, we believe that the high motion accuracy and high dexterity of the fully actuated dexterous hand will facilitate the completion of more complex robotic grasping tasks.

Some studies have designed reconfigurable palm mechanisms and found that reconfigurable palm mechanisms can increase the dexterity of dexterous hands interacting with objects of various shapes and sizes. For example, Wei *et al.* (2011) designed a dexterous hand based on the metamorphic principle. The palm can be deformed, which increases the overall workspace of the dexterous hand. The two fingers of Barrett Hand (Townsend, 2000) can rotate 180° around the central axis of the palm, forming a variety of grasping postures. Lu *et al.* (2021) designed a five-link mechanism located on the palm of the dexterous hand, using two actuators to control the reconstruction of the palm. The addition of reconfigurable palms to the dexterous hand system introduces multiple grasping modes, simplifying the mechanical structure while enabling effective dexterous manipulation.

Force perception plays an important role in robot grasping and manipulation. To enhance stable grasping ability, researchers have installed tactile sensors on dexterous hands, enabling the acquisition of force information during external contact. For example, Wettels *et al.* (2009) developed BioTAC sensors to develop force control on the hands of Otto Bock robots. Lu integrated tactile sensors into the distal phalanx and middle phalanx of each finger and the palm of the dexterous hand (Lu *et al.*, 2022). These studies have shown the effectiveness of using force information to improve dexterous hand grasping. Using a six-axis force sensor instead of the traditional tactile sensor, the dexterous hand can obtain more rich force information such as object position, friction and subtle differences in surface roughness. For example, DLR hand (Butterfass *et al.*, 1998), Gifu hand (Kawasaki *et al.*, 2002), DLR/HIT II hand (Liu *et al.*, 2008), etc. adopt six-axis force sensors installed at the finger to facilitate grasping or manipulation of dexterous hand. By using more

multidimensional force information feedback, it can bring higher dexterity and safety to the dexterous hand and enhance the application ability of the dexterous hand.

The various types of dexterous hands mentioned above have demonstrated remarkable advancements in dexterous grasping, robust manipulation, sensor integration and mechanical structure design. However, it is challenging for a single dexterous hand to encompass all of these advantages simultaneously. To solve the problems above, a multi-configuration dexterous hand integrated with high-dimensional sensors was presented in previous work (Chen *et al.*, 2023). The dexterous hand has nine DOF and adopts a reconfigurable palm design, which can switch different grasping postures. At the same time, each finger of the dexterous hand is equipped with two embedded six-axis force/torque (F/T) sensors, which makes it have stronger sensing ability. The whole dexterous hand is simple in structure and can realize multi-dimensional perception and dexterous manipulation. In this paper, we mainly focus on the grasping ability verification based on kinematics analysis: workspace, interactivity of fingers and manipulability. Moreover, the related experiments were carried out, including finger performance experiments, grasping diversity, in-hand manipulation ability and comparison, to support the effectiveness of the design method.

The rest of this paper is organized as follows. Section 2 introduces the design of the dexterous hand and its modules. Section 3 carries out kinematics modeling and kinematics performance analysis of the dexterous hand. Section 4 evaluates the finger ability test, grasping and in-hand manipulation ability of the dexterous hand and compares it with other dexterous hands. Finally, Section 5 summarizes the paper.

2. Dexterous hand system design

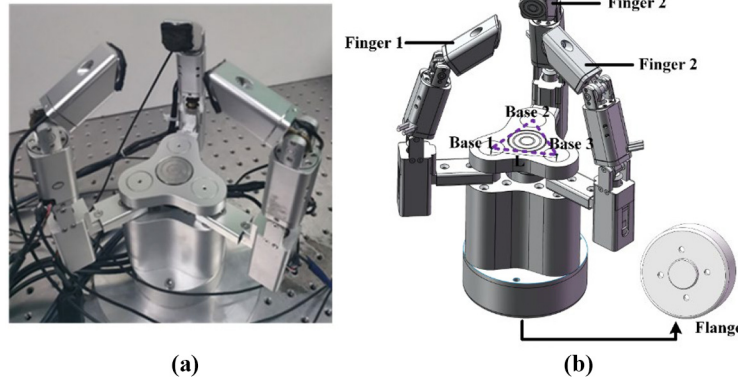
2.1 Structure design

The proposed three-fingered dexterous hand comprises a palm and three individual fingers, as shown in Figure 1(a). Three fingers have the identical modular structure, and each finger has three rotational DOF corresponding to three finger joints. Thus, the entire dexterous hand has a total of nine DOF, directly driven by nine motors through gear transmission. The basic parameters of the dexterous hand are shown in Table 1.

2.1.1 Palm structure design

The layout of the fingers will have a significant impact on the workspace and grasping ability of the dexterous hand (Hota and Kumar, 2019). Therefore, for the needs of grasping, this paper designs a reconfigurable dexterous palm that can change the position of fingers. The three fingers are evenly and symmetrically distributed on the palm, located at the three vertices of the equilateral triangle, as shown in Figure 1(b). The rotation axis between the palm and the proximal interphalangeal is perpendicular to the palm plane, which allows each finger to rotate to adjust its position. As a result, the dexterous hand can adopt a variety of grasping modes, enabling the completion of various challenging grasping tasks.

By adjusting the configuration of the three fingers, four typical grasping modes can be obtained, as shown in Figure 2(a)–(d). As shown in Figure 2(a), tripod precision grasp and

Figure 1 Three-fingered dexterous hand

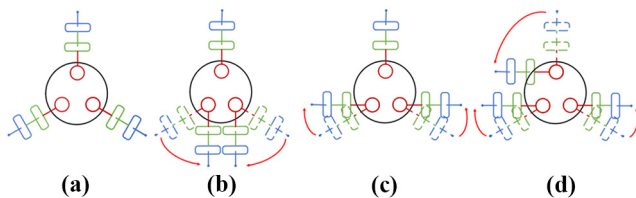
Notes: (a) Prototype; (b) 3D model
Source: Authors' own work

Table 1 Parameters of dexterous hand

Quantity	Value
Size	185 × 185 × 368 mm ³
DOF	9
Number of actuators	9
Payload	4.5 kg
Communication	Ethernet
Rated voltage	24V

Source: Authors' own work

spherical grasp can be achieved when the three fingers are distributed at 120°. As shown in Figure 2(b), when two fingers are rotated together to form a “one finger,” and the position of the third finger is opposite, cylindrical grasp and parallel grasp can be realized. As shown in Figure 2(c), when the two fingers rotate to the relative position, the two fingers can realize the traditional two fingers clamping. The third finger rotates to a vertical position with these two fingers, and the three fingers form a T-shaped grasp posture, which is equivalent to combining the first two grasp postures together to achieve a wider range of grasp. As shown in Figure 2(d), two fingers rotate to the opposite position, and the third finger rotates to be parallel to either of the two fingers, which can realize the hand pinching action.

Figure 2 Four different modes of grasp

Notes: (a) Tripod grasp; (b) parallel grasp; (c) T-shaped grasp; (d) pinch grasp

Source: Authors' own work

In addition, we also design a flange compatible with the robotic arm, as shown in Figure 1(b). Through the flange, the dexterous hand can be directly installed on the robotic arm without changing the body structure.

2.1.2 Finger structure design

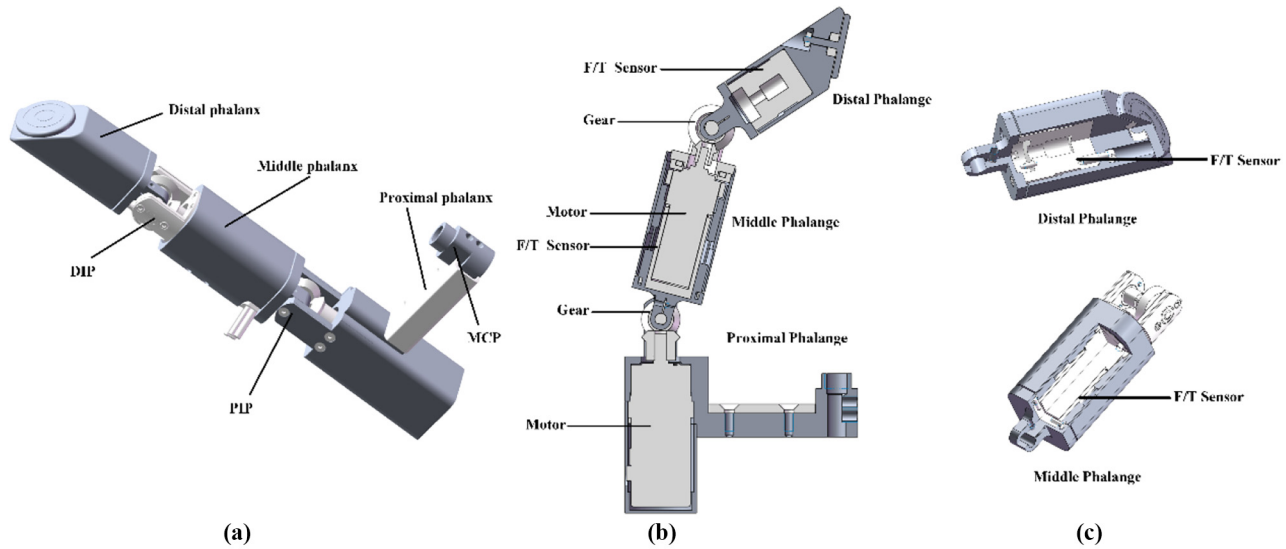
A modular design approach is implemented for the dexterous hand, wherein each finger possesses an identical structure to achieve a high degree of modularity. The finger drives, gears, etc. are integrated directly into the palm and inside the finger phalanx. Inspired by the versatility of the human hand and corresponding to human anatomy, each finger module of this dexterous hand has three phalanxes, including the proximal, middle and distal phalanxes, and three joints, including the distal interphalangeal joint (DIP), proximal interphalangeal joint (PIP) and metacarpophalangeal joint (MCP), as shown in Figure 3(a). The range of motion of DIP, PIP and MCP is -45° to 90° , -90° to 90° and -120° to 120° , respectively.

To reduce the complexity of the transmission mechanism and make full use of space, space is reduced by placing the motor vertically in the phalanx. The bevel gear differential transmission is used as the transmission mode between adjacent knuckles. Each joint has a pair of spiral bevel gears with a transmission ratio of 1:2, as shown in Figure 3(b). The transmission mechanism is realized by bevel gears. The driving wheel is installed on the rotation axis of the finger motor, and the driven wheel side is placed at the finger joint connector. To reduce the transmission error and improve system reliability, the motion uncertainty caused by gear clearance is reduced by high-precision machining and assembly. For the motion error of the dexterous hand, we will get accurate results through an experiment in Section 4.1.

2.2 Integration of the high-dimensional sensors

It is difficult to achieve stable grasping and manipulation only by position control. Especially for irregular objects, the grasping effect is more dependent on the accuracy and dimension of force perception. In addition, the installation of sensors on the finger will affect the volume and operational dexterity of the finger.

In terms of above problems, we designed two six-axis F/T sensors integrated in the distal and middle phalanxes of each

Figure 3 Finger 3D model and sectional drawing

Notes: (a) 3D model of a finger; (b) overall sectional drawing of the finger; (c) partial sectional drawing of the phalanx
Source: Authors' own work

fingers, as shown in Figure 3(c). The six-axis F/T sensors can measure forces and moments along three orthogonal axes in three-dimensional (3D) space, thereby providing effective force information for controlling the dexterous hand. The designed six-axis F/T force sensor has a diameter of 18 mm, a length of 23 mm, a force range of 10 N and a torque range of 0.06 N·m. The F/T sensor is designed using a resistance strain gauge, and three sets of strain gauges are arranged in the area shown in Figure 4. The elastomer deforms under the action of external force. The strain gauge attached to the elastomer produces a strain proportional to the magnitude of the external force and then converts the strain into an electrical signal output through a circuit.

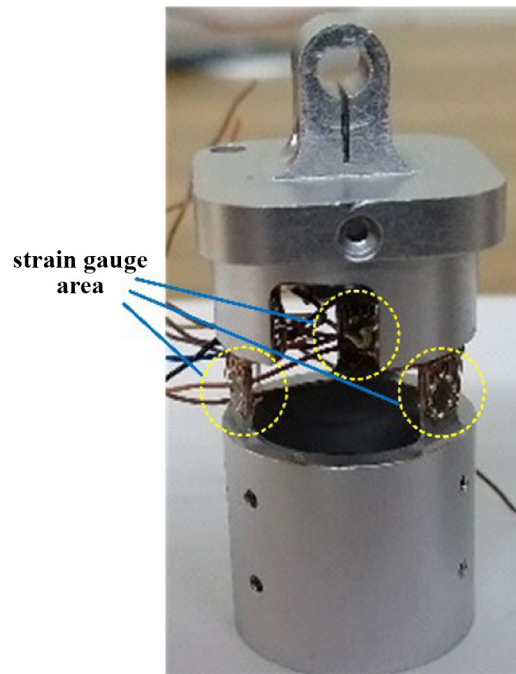
3. Kinematics analysis of dexterous hand

3.1 Kinematic model

3.1.1 Forward kinematics analysis

Kinematics analysis is to obtain the position and orientation of the fingertip in the workspace by establishing the homogeneous coordinate transformation relationship between the finger linkages to obtain the pose matrix of the fingertip. Each joint of the dexterous hand is independently controlled, which means that the correct dexterous hand kinematics need to be established to meet the control requirements, and the establishment of kinematic relationships between the fingers is essential for obtaining workspace.

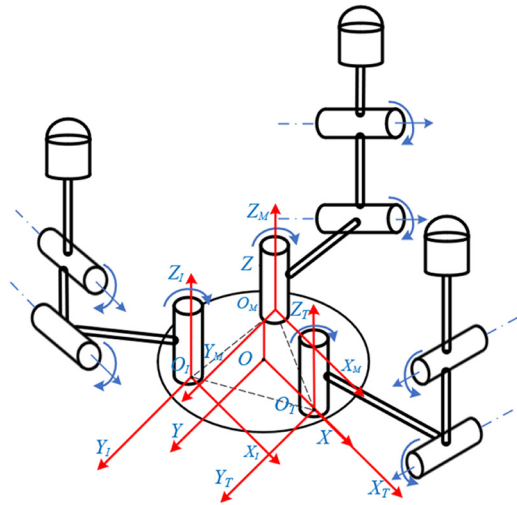
The coordinate system of the dexterous hand (including the palm) is established based on the structure model of the dexterous hand. According to the relationship between the fingers on the palm, the DH method is used to establish the coordinate system, as shown in Figure 5. In the figure, $OXYZ$ is the palm coordinate system established at point O , $O_T X_T Y_T Z_T$ is the thumb coordinate system, $O_I Z_I X_I Y_I$ is the index finger coordinate system and $O_M X_M Y_M Z_M$ is the middle finger coordinate system, where the thumb coordinate system is on

Figure 4 Hardware design of the proposed F/T sensor

Source: Authors' own work

the OX axis. For example, the D-H parameters of the index finger joint are shown in Table 2.

The kinematic relationship of the three fingers of the three-finger dexterous hand is the same. According to the kinematics of the finger, the coordinate transformation matrix of the fingertip relative to the finger base coordinate system at the

Figure 5 Coordinate systems of the dexterous hand

Source: Authors' own work

Table 2 D-H parameters

Joint	α_i	a_i	d_i	θ_i
1	0	30.3	0	0
2	$\pi/2$	77.8	0	θ_2
3	0	87.86	0	θ_3
4	0	49.5	0	θ_4

Source: Authors' own work

base joint of each finger can be obtained. According to the palm coordinate system, the coordinate transformation matrix of the index finger, middle finger and thumb base coordinate system relative to the palm coordinate system can be obtained. By using the coordinate transformation matrix of each fingertip relative to the finger base coordinate system and the coordinate transformation matrix of the finger base coordinate system relative to the palm coordinate system, the transformation matrix of each fingertip relative to the palm coordinate system (the reference coordinate system) can be obtained. List the coordinate transformation matrices between the joints:

$$\begin{aligned}
 {}^0_1A(\theta_1) &= \begin{bmatrix} c_1 & 0 & s_1 & a_1c_1 \\ s_1 & 0 & -c_1 & a_1s_1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \\
 {}^1_2A(\theta_2) &= \begin{bmatrix} c_2 & -s_2 & 0 & a_2c_2 \\ s_2 & c_2 & 0 & a_2s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \\
 {}^2_3A(\theta_3) &= \begin{bmatrix} c_3 & -s_3 & 0 & a_3c_3 \\ s_3 & c_3 & 0 & a_3s_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (1)
 \end{aligned}$$

From the above formula, $s_i = \sin \theta_i$, $c_i = \cos \theta_i$, multiply the obtained joint transformation matrix to obtain the total coordinate transformation matrix of the fingertip relative to the finger base coordinate system, expressed as:

$$\begin{aligned}
 {}^0_3T(q) &= {}^0_1A_2^1A_3^2A \\
 &= \begin{bmatrix} c_1c_{23} & -c_1s_{23} & s_1 & c_1(a_1 + a_2s_2 + a_3c_{23}) \\ s_1c_{23} & -s_1s_{23} & -c_1 & s_1(a_1 + a_2c_2 + a_3c_{23}) \\ s_{23} & c_{23} & 0 & a_2s_2 + a_3s_{23} \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (2) \\
 q &= [\theta_1 \quad \theta_2 \quad \theta_3]^T
 \end{aligned}$$

where $c_{ab} = \cos(\theta_a + \theta_b)$ and $s_{ab} = \sin(\theta_a + \theta_b)$. The above formula is the kinematic model of the index finger. The three fingers are positioned equidistantly, with each finger originating from one vertex of an equilateral triangle configuration. Thumb base coordinate system, index finger base coordinate system and middle finger base coordinate system are, respectively, located at 90° , 120° and 240° of OX axis. The coordinate transformation matrix of the finger base coordinate system relative to the palm coordinate system can be obtained:

$${}^b_0A(\theta_0) = \begin{bmatrix} c_0 & -s_0 & 0 & a_0c_0 \\ s_0 & c_0 & 0 & a_0s_0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (3)$$

By multiplying with equation (2), the coordinate transformation matrix of the fingertip relative to the palm coordinate system can be obtained:

$$T = {}^b_3T(q) = {}^b_0A_1^0A_2^1A_3^2A = \begin{bmatrix} R & P \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (4)$$

where R is the rotation matrix and P is the position matrix.

3.1.2 Inverse kinematics analysis

The inverse kinematics solution aims to solve the rotation angle variable of each finger joint under the premise of the given position of the fingertip. Suppose that the pose transformation matrix of the fingertip in the base coordinate system is expressed as:

$$Q = \begin{bmatrix} n_x & o_x & a_x & x \\ n_y & o_y & a_y & y \\ n_z & o_z & a_z & z \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (5)$$

When $Q = T$, the inverse kinematics of the finger can be obtained:

$$\theta_1 = \arctan\left(\frac{yc_0 - xs_0}{xc_0 + ys_0 - a_0}\right) \quad (6)$$

$$c_3 = \frac{k^2 + z^2 - a_2^2 - a_3^2}{2a_2a_3} \quad (7)$$

where $k = xc_0c_1 - xs_0s_1 + yc_0s_1 + yc_1s_0 - a_0c_1 - a_1$.

$$s_3 = \pm \sqrt{1 - c_3^2} \quad (8)$$

$$\theta_3 = \text{Atan}(s_3, c_3) \quad (9)$$

$$\theta_2 = \arcsin\left(\frac{z}{\sqrt{a_2^2 + a_3^2 + 2a_2a_3c_3}}\right) - \arctan\left(\frac{a_3s_3}{a_2 + s_3c_3}\right) \quad (10)$$

According to equations (6), (9) and (10), the rotation angle variables of each joint of the dexterous hand can be obtained by obtaining the pose of the fingertip.

3.2 Performance analysis of dexterous hand

To evaluate the performance of dexterous hands, some important criteria for evaluating the performance of dexterous hands can be derived from relevant literature (Kashef et al., 2020). Based on these criteria, we select three key indicators to evaluate the dexterous hand proposed in this paper: workspace, interactivity of fingers and manipulability.

3.2.1 Workspace

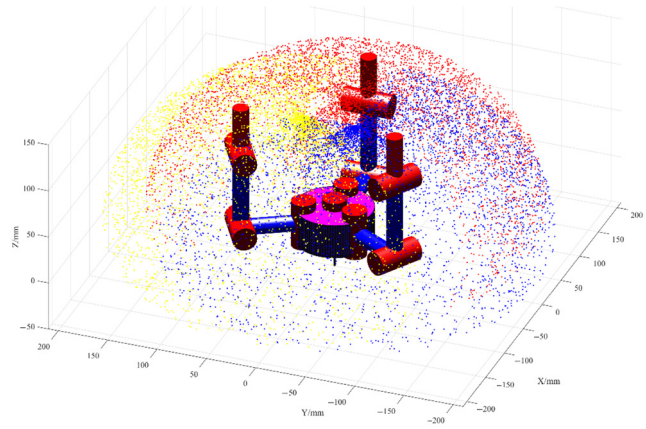
The workspace is one of the important performance metrics for dexterous hands, as it intuitively characterizes the movement ability, performance, accuracy and dexterity. The workspace of the dexterous hand is determined by the number of fingers, joint length, range of motion, spatial layout and other parameters and conditions. Establish the workspace of the dexterous hand, which is the total spatial range that the fingertips can reach when combining all possible joint configurations. Using MATLAB software, the workspace point cloud of the dexterous hand can be graphically represented, as shown in Figure 6. The total workspace volume of the dexterous hand and the workspace volume of the interaction between the fingers can then be calculated. The total workspace volume of a dexterous hand is the union of three finger workspaces:

$$V = v_1 \cup v_2 \cup v_3 \quad (11)$$

where V is the total workspace volume of the dexterous hand and v_i is the workspace volume of Fingers 1, 2 and 3.

To effectively calculate the workspace volume, we use the Monte–Carlo method. First, the joint angle variables are generated by the Monte–Carlo method, and then the finger joint angle variables are input into the forward kinematics model of the dexterous hand to calculate the sampling points of each fingertip. A bounding box is established for the largest X , Y and Z coordinates in the set of sampling points, and a series of cells are obtained by slicing in the X , Y and Z directions; each cell volume is ΔV . Then, the sampling points are marked in the

Figure 6 Workspace simulation of the dexterous hands



Source: Authors' own work

bounding box, and n is the number of marked cells. Therefore, the workspace volume of each finger of the dexterous hand is computed as:

$$v_i = n_i \Delta V \quad (12)$$

After calculation, the volume of the intersecting workspace of each finger can also be calculated by the above formula. The volume of the intersecting workspace of the two fingers is $317,167 \text{ mm}^3$. Therefore, the dexterous hand designed in this paper has good operability and practicability.

3.2.2 Interactivity of fingers

The interaction of multiple fingers gives the hand a complex workspace and the ability to grasp various objects. The human hand exhibits exceptional dexterity and versatility in grasping owing to the overlapping workspace between the fingers.

You et al. (2019) proposed a new dexterous hand performance index called “interactivity of fingers (IF).” It objectively evaluates the performance of the dexterous hand by using the intersection volume of the workspace between the thumb and other fingers, and the intersection volume of the workspace between the thumb and two adjacent fingers. The definition of finger interactivity is as follows:

$$IF = \frac{\sum_{i=1}^{n-1} \mu_i d_i}{(n-1)\theta_{total} L_t^2 L_f^2} \left(\sum_{i=1}^{n-1} \mu_i v_i + k \sum_{i=1}^{n-2} \mu_i \mu_j v_{ij} \right) \quad (13)$$

where d_i is the distance between the base of the thumb and the MCP joint position of the i th finger, L_t is the length of the thumb, L_f is the length of the index finger to the little finger, θ_{total} represents the sum of the motion range of the controllable joint, n is the number of fingers including the thumb, v_i represents the workspace volume of the intersection of the thumb and the i th finger, v_{ij} is the workspace volume of the intersection of the thumb and two adjacent fingers and μ_i is the weighting coefficient. The value of i is 1–4; each number represents the index finger to the little finger; and the value of j ($= i + 1$) is used to represent the finger adjacent to the i th finger. The weighted coefficient is defined as $\mu = [1, 1, 0.5,$

0.5]. Because the value of v_{ij} is inevitably less than v_i , the compensation coefficient (k) maintains the balance of the influence of v_i and v_{ij} on IF .

This performance index can be used to determine the optimal kinematic configuration of the finger. The finger interaction values of the dexterous hand are calculated using the above definitions. The final results and the finger interaction values of some existing dexterous hands are listed in Table 3. The comparison shows that the dexterous hand proposed in this paper has good finger interaction. Although the finger interaction ability of the proposed hand is lower than that of the four- or five-finger dexterous hand because of the DOF, the dexterous hand proposed in this paper has a higher finger interaction ability than the traditional three-finger dexterous hand. This again proves that the dexterous hand proposed in this paper has excellent kinematic configuration because of the design of our finger number, joint length and range of motion.

3.2.3 Manipulability

Although finger interaction can effectively quantify the performance of dexterous hand, it cannot provide information about the dexterity of dexterous hand in certain postures or tasks. In robotics, manipulability is defined as the ability to move and exert force in any direction (Angeles and Park, 2008). This definition means that manipulability depends on the robot specific posture. In this section, by calculating the manipulability of the dexterous hand, the concept of kinematic manipulability is used to analyze the dexterity of the dexterous hand.

The finger Jacobian matrix of a dexterous hand can be defined as a linear transformation between its operating speed and joint speed, which is considered as a transmission ratio from joint space to operating space. Set the motion equation of the finger to:

$$\dot{x} = J(q)\dot{q} \quad (14)$$

Equation (14) defines the kinematic mapping between the task space x and the joint space q . By differentiating both sides of the formula with respect to time t , the differential relationship between q and x can be obtained:

$$\dot{x} = J(q)\dot{q} \quad (15)$$

Thus, the Jacobi matrix of the dexterous hand finger can be derived as:

Table 3 Interactivity of fingers

Type	No. fingers	IF [(10^{-4})]
DLR-Hand	5	7.2055
SKKU Hand IV (Kim et al., 2014)	4	4.5039
UB Hand 4 (Melchiorri et al., 2013)	4	4.1942
Our hand	3	2.0841
Schunk SDH hand	3	1.7258
Barrett Hand (Townsend, 2000)	3	1.4306

Source: Authors' own work

$$J(q) = \begin{bmatrix} -s_1(a_1 + a_2s_2 + a_3s_{23}) & c_1c_{23}(a_3 + a_2c_3) + a_2c_1s_3s_{23} & a_3c_1c_{23} \\ c_1(a_1 + a_2s_2 + a_3s_{23}) & s_1c_{23}(a_3 + a_2c_3) + a_2s_1s_3s_{23} & a_3s_1c_{23} \\ 0 & s_{23}(a_3 + a_2c_3) - a_2s_3c_{23} & a_3s_{23} \end{bmatrix} \quad (16)$$

Manipulability is an effective indicator of the dexterity of a robot in 3D space. It was proposed by Yoshikawa (1985) and is defined as follows:

$$w = \sqrt{\det(J(q)J(q)^T)} \quad (17)$$

In the formula, the Jacobian matrix of the finger at the pose q is represented as $J(q)$. Because the Jacobian matrix is a function of the finger pose, for a redundant manipulator, there is an infinite number of manipulability at any point in the space. The fingers of our dexterous hand can be regarded as redundant manipulators in 3D space. Therefore, it is not easy to visualize the distribution of manipulability in 3D space. To effectively visualize the manipulability, we normalize it to make its dimension unified. The normalized manipulability is defined as follows:

$$w_n = \frac{w_i}{\max(w_1, w_2, w_3, \dots, w_n)} \quad (18)$$

where w_i is the manipulability value of the given point in the workspace; $\max(w_1, w_2, w_3, \dots, w_n)$ is the maximum manipulability value in the whole workspace.

The singularity of the robot can be directly identified by the manipulability index. The manipulability index is between [0, 1]. When the manipulability of the robot is 0, the finger is in a singular configuration, which will reduce its kinematic performance. In practical applications, it is imperative to avoid singular configurations. Observably, the higher the manipulability value, the more away from the singular configuration, with better operating performance. When $w_n = 1$, it means that the point has the best motion performance.

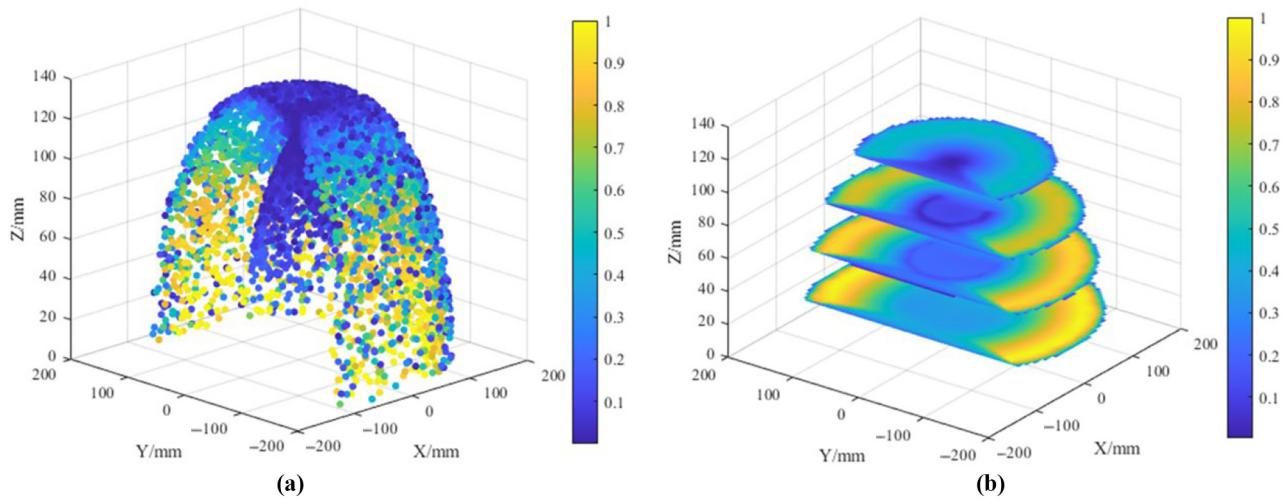
By using 20,000 sampling points, the distribution of the dexterous hand manipulability is obtained, as shown in Figure 7. The areas with high manipulability appear near the center and end of the finger workspace of the dexterous hand. In typical precision grasping tasks, the fingertip is usually located in the middle of the finger workspace. This is in the high manipulability area of our dexterous hand, which means that our dexterous hand has excellent grasping ability. This advantage will be demonstrated in Experiment 2 in Section 4.

4. Experiment and discussion

The basic parameters of the dexterous hand and its grasping and manipulation capabilities are evaluated through experimental verification.

4.1 Fingers performance experiments

Before the grasping evaluation experiment, it is necessary to quantify the performance of the dexterous hand. Its mechanical specifications (finger/hand size, range of motion, etc., as shown in Table 1) do not require experiments to quantify. However,

Figure 7 Manipulability analysis of the dexterous hand

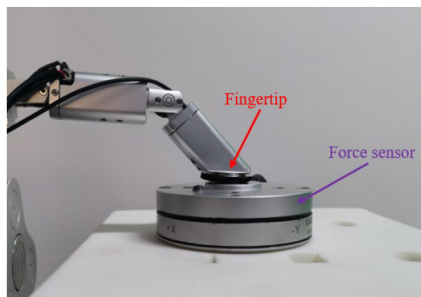
Notes: (a) Points cloud; (b) slice map

Source: Authors' own work

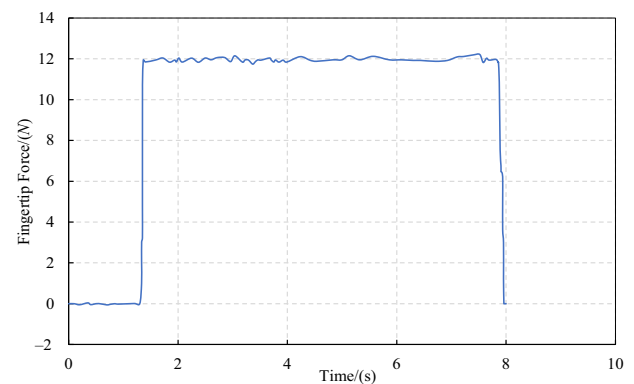
the coefficients of fingertip force and finger motion accuracy still need to be determined by quantitative experiments.

To assess fingertip force, the dexterous hand is fixed to a plane and measured using a reference six-axis force sensor (M4313A; Sunrise Instruments, China) fixed to a plane perpendicular to it, as shown in Figure 8. By controlling the finger bending of the dexterous hand, and pressing the surface of the six-axis force sensor to exert external force, the force information collected by the sensor data is transmitted to the PC for display. Through experiments, it is found that when a stable voltage of 24 V is provided, the finger of the dexterous hand can output a peak fingertip force of up to 12 N, as shown in Figure 9. The performance test of fingertip force shows that the proposed hand has great potential to realize force control when performing grasping tasks.

For the verification of the kinematics of the dexterous hand, we compared the distance of the fingertip position movement with the fingertip movement distance calculated by the kinematic parameter model. The absolute accuracy of the robot is measured by the distance accuracy between two points in space to avoid the error caused by the transformation matrix between the measurement coordinate system and the robot

Figure 8 Fingertip force experimental setup

Source: Authors' own work

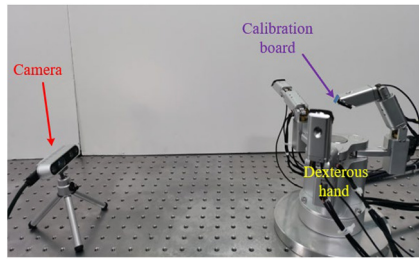
Figure 9 Measured fingertip force data

Source: Authors' own work

base coordinate system. For the detection of robot motion, laser tracker and three-coordinate measuring machine are generally used for measurement. However, to reduce the measurement cost and operation complexity, a large number of researchers have proposed the method of visual measurement (Chen *et al.*, 2014).

In this experiment, monocular vision measurement is used, as shown in Figure 10. The 10×10 mm calibration board of Halcon is mounted as a target on the end of a fingertip. The camera is fixed next to the dexterous hand to visually track the calibration plate and detect the corner information on the calibration board. The Perspective-n-Point algorithm is used for pose measurement. The position change of the mark point of the target is measured, and the moving distance of the fingertip is replaced by the measured distance.

In the experiment, the first point is selected as the starting point of other points, and then 20 measurement points are evenly selected on the space ball (radius of 30 mm) with this point as the center. In each experiment, 20 error values are

Figure 10 Motion accuracy test experimental setup

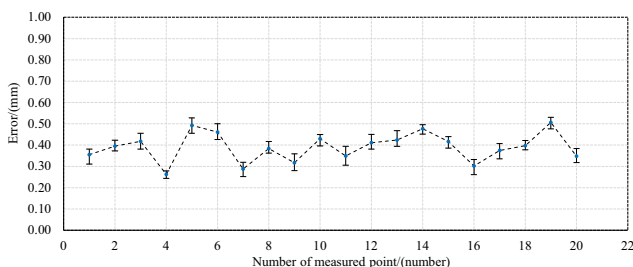
Source: Authors' own work

measured and the average value is calculated. The average motion error obtained in the final experiment is shown in Figure 11. It can be seen from the figure that the maximum average error does not exceed 0.506 mm. The result demonstrates that the kinematic model of the dexterous hand is established correctly and has high control accuracy.

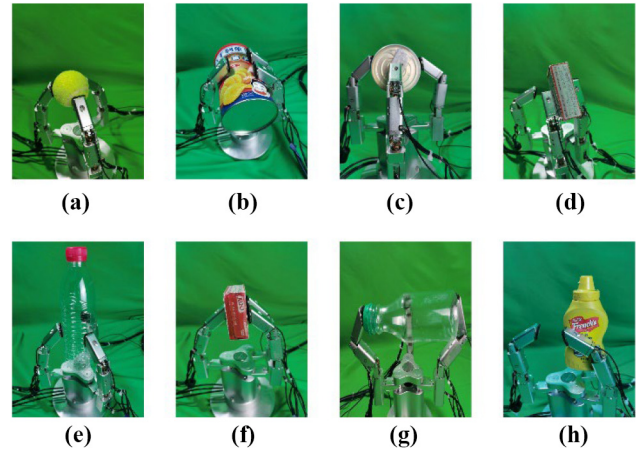
4.2 Grasping experiments

To validate the grasping ability proposed in Section 2.1, eight daily objects are adopted as the experimental subjects. In Figure 12(a) and 12(e), the hand grasps the ball and the water bottle in terms of a tripod grasp gesture. It shows that the dexterous hand can grasp spherical objects and cylindrical objects stably through the tripod grasp mode. In Figure 12(b) and 12(f), hand grasps the chips barrel and the box in terms of a parallel grasp gesture. It shows that the dexterous hand can grasp cylindrical objects and square objects stably through the parallel grasp mode. In Figure 12(c) and 12(g), hand grasps the can and the water bottle in terms of a T-shaped grasp gesture. It shows that the dexterous hand can grasp cylindrical objects stably through the T-shaped grasp mode. In Figure 12(d) and 12(h), hand grasps the box and the mustard bottle in terms of a pinch grasp gesture. It shows that the dexterous hand can grasp square objects and irregular objects stably through the pinch grasp mode. The experimental results reveal that the dexterous hand can perform various grasping tasks.

To further evaluate the dexterity of the three-fingered dexterous hand in grasping tasks, the grasping test of the dexterous hand is carried out based on the human grasping taxonomy proposed by Feix *et al.* (2016). In Feix taxonomy, 33 commonly used human hand grasping methods are systematically categorized. Not all grasping types require the use of all five fingers. Some common grasping postures, such as

Figure 11 Average error curve of motion accuracy test

Source: Authors' own work

Figure 12 Dexterous hand grasps various daily objects relying on four typical grasping modes

Source: Authors' own work

spherical grasping, triangular grasping and pinching, can be achieved using only four, three or even two fingers of the human hand. Additionally, some grasping postures, such as spherical grasping and precise grasping, can be completed using three, four or all five fingers. Therefore, our experimental goal is to perform static evaluation by using a three-fingered dexterous hand to complete as many grasping postures listed in the Feix taxonomy as possible.

Finally, after testing experiments, we successfully realized 27 types of human hand grasping in all 33 types of Feix taxonomy on the three-finger dexterous hand. The experimental results are shown in Figure 13. The six types of grasping that are not implemented require four or five fingers. Because we design a three-finger dexterous hand, these six grasping types cannot be achieved.

The experimental results reveal that the proposed three-finger dexterous hand can replicate 82% of the natural movement of the human hand in daily life grasping activities by adjusting the finger configuration gesture. Therefore, from a practical point of view, the proposed hand can mimic the versatile functionality of the human hand and has potential application in service robots and industrial automation tasks.

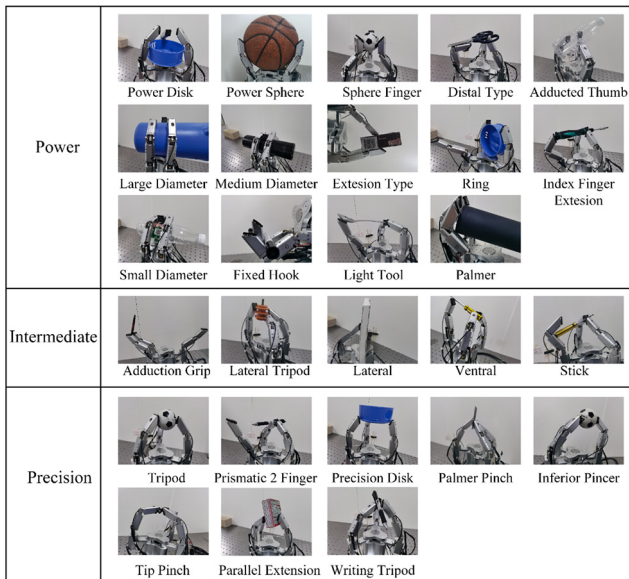
4.3 In-hand manipulation experiments

In practical applications, the dexterous hand not only performs simple grasping but also adjusts the posture of the object in the hand to reposition according to the task. Therefore, this section verifies the in-hand manipulation ability of the dexterous hand through the operation test of the in-hand object.

In Figure 14(a), the cylinder is reoriented to the other side of the finger. In Figure 14(b), the object is repositioned from the left side of the finger to the right side. In Figure 14(c), the dexterous hand translates the object vertically upward. In Figure 14(d), the dexterous hand uses a relative two-finger grasp, and then rotates and turns into a firmer spherical triangle grasp. Experiments have shown that dexterous hands can perform several in-hand manipulation tasks.

Through simple position control, the dexterous hand can perform certain plane or space manipulations on the grasped

Figure 13 Grasp poses performed by the proposed hand according to Feix taxonomy



Source: Authors' own work

object. The results show that the dexterous hand has preliminary dexterous in-hand manipulation performance.

The sensors integrated in the hand have not yet been applied to a variety of manipulation tasks. This needs to be addressed in future work as part of the force control algorithm to achieve more intelligent control.

4.4 Comparison

In comparison to existing dexterous hands, the results are presented in Table 4, which compares various aspects, including the weight, the number of actuators and fingers, size,

perception and more. Ours hand use full drive and gear transmission, which makes the hand load capacity greater. Compared with the classic Barrett Hand, the dexterous hand of this paper has more and higher dimensional sensors, which means that the dexterous hand we proposed has stronger perception ability. In terms of kinematics, the dexterous hand of this paper has one more DOF than Barrett Hand, which means that the dexterous hand we proposed can perform richer grasping postures. The experiment in Section 4.2 also proves this.

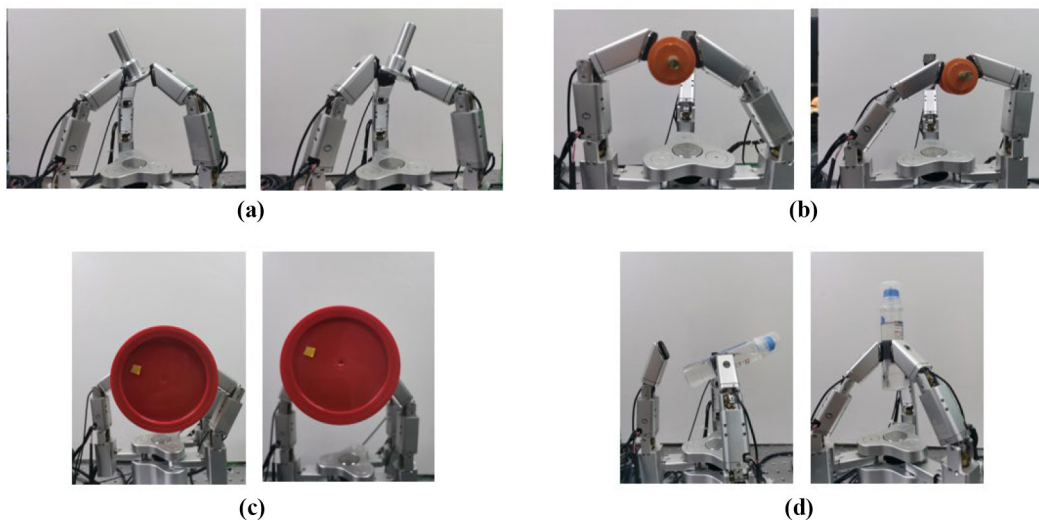
The dexterous hand designed in this paper has rich perception ability, more actuators, a large grasping range and suitable weight. The dexterous hand of this design has higher DOF and actuators, allowing it to perform complex operational tasks. Force feedback can also help the dexterous hand to achieve grasping and in-hand manipulation. The use of a reconfigurable palm allows the proposed dexterous hand to perform a wide range of grasping and in-hand manipulation tasks while avoiding the common problems of high weight and complex control of high-function humanoid dexterous hands.

5. Conclusion

This study presents a multi-configuration dexterous hand with integrated high-dimensional sensors. The hand consists of a palm and three fingers. Six-axis force sensors and motors are integrated into the phalange links via a modular design approach. The reconfigurable palm enables the hand to adopt four grasping configurations to meet the requirements of various tasks.

Furthermore, the kinematic model of the dexterous hand is established. The performance of the proposed dexterous hand is evaluated through simulation and experimental tests. In simulation, the workspace, interactivity of fingers and manipulability are used to assess the hand's performance. The results demonstrated that the proposed hand has high dexterity. Experimental tests were also conducted to assess the

Figure 14 In-hand manipulations



Source: Authors' own work

Table 4 Dexterous hand comparison

Hand	Fingers	Actuators	DOF	Sensors	Size (mm)	Weight (kg)
Schunk SDH	3	7	7	6-Force sensors, 7-position sensors	86.5/68.5/66	1.95
Robotiq 2-finger	2	1	1	–	90/86/140	1
Robotiq 3-finger	3	2	3	–	126/100/126	2.3
Eagle shoal (Wang et al., 2019)	3	8	8	16-Force sensors	57/71/98	0.79
Kinova KG-3	3	3	6	–	46/46/160	0.68
Barrett Hand (Townsend, 2000)	3	4	8	3-Force sensors, 4-position sensors	70/58/100	1.2
Our hand	3	9	9	6-Axis F/T sensors, 9-position sensors	185/185/368	2.1

Source: Authors' own work

fingertip force motion accuracy, grasping ability and in-hand manipulation of the dexterous hand. The fingertip force test showed that the hand can generate up to 12N controllable fingertip force. The motion accuracy test results revealed that the dexterous hand has high control accuracy, validating the established kinematic model. Based on Feix taxonomy, 27 out of the 33 types of hand grasping in daily life were realized, demonstrating the acceptable grasping ability. Additionally, by controlling the rotation, translation and repositioning of the objects, the dexterous hand demonstrated preliminary in-hand manipulation capability.

In this study, we have not yet carried out experiments or applications for this high-dimensional sensor embedded in the fingers. However, we acknowledge the significance of this area for our future research endeavors. Specifically, as part of our future work, we aim to develop a learning-based controller that uses rich force feedback information for the more complex grasping and dexterous manipulation tasks.

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